

Shamvi Md Abdullah

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GitHub | LinkedIn | Google Scholar | ORCID

SUMMARY

Final-year Computer Science & Engineering student whose work spans machine learning, NLP, computer vision, graph theory, human-computer interaction and data science. I enjoy working in a team, take deadlines seriously, and am happy to learn whatever a project asks of me. Looking for a machine learning, data or research internship to gain hands-on experience in a professional environment.

EDUCATION

American International University-Bangladesh <i>BSc in Computer Science & Engineering, Major in Computational Theory</i> - Expected January 2027. Minor in Data Science. - Coursework: Machine Learning, NLP, Artificial Intelligence, Computational Statistics, Databases.	Dhaka, Bangladesh <i>CGPA: 3.96/4.00</i>
Scholar's School & College <i>Higher Secondary Certificate (Science)</i>	Dhaka, Bangladesh <i>GPA: 5.00/5.00</i>
Scholar's School & College <i>Secondary School Certificate (Science)</i>	Dhaka, Bangladesh <i>GPA: 5.00/5.00</i>

PROJECTS

- MedEase BD** | *Python, NLP, RAG, LLM Fine-Tuning, Flask*
- Developed a trilingual medicine information assistant handling English, Bangla and Banglish queries using Retrieval-Augmented Generation over a corpus of 21,000+ medicines and 1,700+ generic drug records.
 - Fine-tuned a Gemma 3 4B language model on 36,000+ medicine-related question-answer pairs built from Bangladeshi pharmaceutical datasets, and deployed the system to run fully offline.
- Gesture Controlled Robotic Hand** | *Python, MediaPipe, OpenCV, ESP32, UDP*
- Developed a camera-only gesture control system that replicates human hand movement on a robotic hand with no wearable sensors, computing finger joint angles from MediaPipe landmarks to drive seven servos.
 - Achieved 88 ms mean end-to-end latency at 24.6 FPS over a low-latency UDP link to an ESP32 and PCA9685 driver, for a total build cost of about 2,800 BDT.
- Adaptive-Baseline PM2.5 Monitor** | *ESP32-C3, PMS7003, BLE, React Native, Firebase*
- Built a pocket-sized air quality monitor and a React Native app that syncs over BLE and logs spike events to Firebase, for a component cost of about 4,500 BDT against commercial units costing several hundred dollars.
 - Designed an adaptive spike-detection algorithm over a 60-sample rolling baseline (2-sigma threshold, 15 ug/m3 minimum rise, 10-minute cooldown) so alerts fire only on genuine spikes rather than on Dhaka's constantly elevated background.
 - Ran a two-week field study in Dhaka: fixed-threshold alerting produced dozens of alerts per day and was ignored within hours, while adaptive alerting produced roughly four per day, each of which was read and acted on.
- OshudBot** | *Python, Streamlit, RapidFuzz, Docker*
- Built a Bengali medicine assistant covering 21,714 brands that answers queries in Bengali, English or Banglish in roughly 15 ms on CPU, within 330 MB of memory.
 - Measured the contribution of an earlier bangla-bert NER model and multilingual sentence encoder, found the encoder returned confident but wrong matches on Banglish input, and replaced 1.1 GB of models with transliteration and fuzzy matching at no loss in answer quality.
- NeedSurveyResponses** | *PHP, MySQL, JavaScript, HTML, CSS*
- Built a research support platform on an MVC structure where participants earn credits for completing surveys and spend them to publish their own, with session-based authentication and separate user and admin views.
 - Designed the credit system and validation flow, and integrated Google Forms for external surveys while keeping tracking centralised in the database.

PUBLICATIONS

Gesture Controlled Robotic Hand Designed for Enhancing Industrial Automation and Innovation	<i>IEEE QPAIN 2026</i>
Second author and corresponding author. 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking, IEEE Photonics Society. doi:10.1109/QPAIN69676.2026.11545528	
Development of a Simulated Blood-Like Solution for Medical Experiments	<i>Analytical Chemistry Letters, 2025</i>
Fifth author. Taylor & Francis, Vol. 15, Issue 4, pp. 704–717. doi:10.1080/22297928.2025.2533331	
Evaluating the Performance of Agile-Waterfall Integrated Approaches in Large Scale Engineering Projects in Bangladesh	<i>IEOM Bangladesh 2025</i>
Fifth author. 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management. doi:10.46254/BA08.20250467	
A Hybrid Human-AI Model for Sustainable Innovation in Media and Animation	<i>ICCTASS 2025</i>
Third author. Presented at the International Conference on Challenges and Trends in Arts and Social Sciences.	
Development of a Simulated Blood-Like Solution for Medical Experiments	<i>Physics Conference 2024</i>
Presented at the International Conference on Physics 2024, Bangladesh Physical Society, ahead of the journal publication listed above.	

TECHNICAL SKILLS

Languages: Python, R, SQL, C++, Java, JavaScript, PHP
Machine Learning & NLP: PyTorch, scikit-learn, OpenCV, MediaPipe, RAG, LLM fine-tuning, CNN ensembles
Embedded & IoT: ESP32, ESP32-C3, PMS7003, BME280, PCA9685, BLE, UDP, I2C
Tools: Git, Docker, Streamlit, L ^A T _E X, Jupyter

AWARDS & HONOURS

Dean’s List Honor Award Recipient	<i>Fall 2024-2025</i>
American International University-Bangladesh	
Dean’s List Honor Award Recipient	<i>Spring 2023-2024</i>
American International University-Bangladesh	
Academic Scholarship	<i>Ongoing</i>
American International University-Bangladesh	

LANGUAGE PROFICIENCY

IELTS (Academic) - Overall Band Score: 8.0	<i>Listening 9.0 Reading 9.0 Speaking 8.0 Writing 6.5</i>
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ACTIVITIES & INVOLVEMENTS

AI Workshop, U.S. Embassy in Bangladesh	American Center, Dhaka
<i>Selected by the Faculty of Science and Technology to represent AIUB</i>	
Let’s Talk with the UNDP Resident Representative	AIUB, Dhaka
<i>Selected to represent AIUB in a roundtable on youth engagement in the SDGs</i>	
AIUB English Club	AIUB, Dhaka
<i>Organiser and Content Writer, coordinated 10+ events and workshops for 50+ members</i>	
AIUB Poster Presentation Competition	AIUB, Dhaka
<i>Selected participant</i>	

REFERENCES

Dr. Md. Saef Ullah Miah	saef@aiub.edu
<i>Associate Professor & Director, Department of Computer Science</i>	<i>American International University-Bangladesh</i>